Angled and negative work

antocciphysics | TEACHER KEY | CPA

Learning target: Use force direction to determine work’s sign and size.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 3, Day 14; 3 course days later.*
15 m. The constant-Δv straight-line model omits full orbital dynamics.

02. Explain a worked example (worked)
Friction negative; perpendicular force does zero work.

03. Look and explain (model)
Positive, negative, zero respectively.

04. Build the solution (practice)
+24 J,-6 J, net 18 J.

05. Use the idea (reason)
Its parallel component is zero, so it does no instantaneous work; can change direction without changing kinetic energy.

06. Choose a method (mixed) *Review: Unit 3, Day 5; 12 course days later.*
24 kg·m/s right, so the vector sum remains zero.

07. Exit check (exit)
-20 J.

08. STRETCH: Apply it another way (stretch)
Net +12 J; doubled friction work -16 J gives net +4 J.

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01. Bring back an earlier idea (retrieve) *Review: Unit 3, Day 14; 3 course days later.*
$J=7.5$ N·s; $v_f \approx 0.5$ m/s from $mv_i+J=mv_f$.

02. Explain a worked example (worked)
Friction negative; perpendicular force does zero work.

03. Represent and interpret (model)
Positive, negative, zero respectively.

04. Solve and justify (practice)
Rope 86.6 J; friction -20 J; net 66.6 J.

05. Analyze the reasoning (reason)
Its parallel component is zero, so it does no instantaneous work; can change direction without changing kinetic energy.

06. Choose a method (mixed) *Review: Unit 3, Day 5; 12 course days later.*
Other velocity -8 m/s; total $K=480$ J, supplied by released internal energy.

07. Exit check (exit)
-10 J; component opposes displacement.

08. STRETCH: Apply it another way (stretch)
Pull $20\times3\times\cos 60°=30$ J; friction -12 J; sum 18 J. Vertical forces do no work for horizontal displacement.